MILITARY SPECIFICATION

CARTRIDGE, 20MM, ARMOR PIERCING INCENDIARY-TRACER, M52E1

1. SCOPE.

1.1 This specification covers one type of electric primed cartridge for use in Guns, 20MM, Automatic, M39 and M61.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-644 - Visual Inspection Standards and Inspection Procedures for Inspection of Packaging, Packing, and Marking of Small Arms Ammunition
MIL-STD-651 - Visual Inspection Standards for 20MM Ammunition and Components
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

MUNITIONS COMMAND

C7258801 - Classification of Firing Casualties for Case, Cartridge, 20MM, M103 and M103B1
F7258928 - Marking on Shipping and Storage Container, M548 for 20MM Cartridges
F7258941 - Method of Packing 100 Cartridges, 20MM, M50 Series, Except Dummy, in Bulk, in M548 Container

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F7258942 - Method of Packing 200 Cartridges, 20MM, M50 Series, in Bulk, in M548 Container
F7258968 - Method of Packing 100 Cartridges, 20MM, in M12 Links, in M548 Container
F7258969 - Method of Packing 100 Cartridges, 20MM, in M14 Links, in M548 Container
F7258970 - Method of Packing 100 Cartridges, 20MM, in M17 Links, in M548 Container
D7259218 - Cartridge, 20MM, Armor Piercing Incendiary-Tracer, M52E1
IEL 7259218 - Index of Inspection Equipment Lists for: Cartridge, 20MM, API-T, M52E1
D7259237 - Marking on Shipping and Storage Containers M548 for 20MM Cartridges (Functional)

PUBLICATIONS

MUNITIONS COMMAND

AMCR 715-505 - Ammunition Ballistics Acceptance Test Methods, Volume 8, Test Procedures for 20MM Cartridges

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.-The cartridge shall comply with Drawing (Dwg.) D7259218, referenced specifications, and other requirements specified herein.

3.2 First article testing.-Requirements for the submission of first article samples shall be as specified in the contract. (See 6.2)

3.3 Manufacturing process.-All parts and assemblies shall be manufactured by a process approved by the contracting officer, and no deviation from that process shall be made without his prior approval. (See 6.3)

3.4 Projectile extraction.-The force necessary to extract the projectile from the cartridge case shall be not less than 1100 pounds or more than 2600 pounds.

3.5 Electrical resistance, primed case.-The electrical resistance of the cartridge primer shall be not less than 500 ohms, or more than 1,200,000 ohms.

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3.6 Projectile torque.-The projectile shall withstand a minimum of 10 inch pound torque without rotary movement with respect to the cartridge case.

3.7 Velocity.-The average velocity of projectiles of the sample cartridges conditioned at 68 degrees Fahrenheit (°F) to 72°F., shall be 3380 ± 50 feet per second (fps) at 78 feet from the gun muzzle, and the sample standard deviation shall not exceed 40 fps.

3.8 Pressure.-The average chamber pressure ($\bar{X}$) of the sample cartridges, conditioned at 68°F. to 72°F., plus 0.84 times the average range ($\bar{R}$) of the chamber pressure shall not exceed 60,500 pounds per square inch (psi).

3.9 Waterproofness.-The average velocity (wet) of projectiles of the sample cartridges conditioned at 68°F. to 72°F., shall not vary from the average velocity (dry) by more than 100 fps.

3.10 Accuracy.-The average mean radius of all targets of the sample cartridge projectiles, shall not exceed 15 inches at a range of 600 yards.

3.11 Action time.-The action time of the cartridge shall not exceed 4.0 milliseconds at either ambient or -65°F. temperature.

3.12 Incendiary performance.-The projectile shall function with acceptable incendiary characteristics upon target impact.

3.13 Drop sensitivity.-The projectile shall not function upon impact against steel plate at a velocity of 52 ± 5 fps.

3.14 Projectile trace.-The projectile shall be tracing brightly at a distance not greater than 100 yards from the gun muzzle, and shall continue tracing brightly for a distance not less than 1500 yards from the gun muzzle.

3.15 Cyclic rate.-The cartridges shall operate the M39 gun at a cyclic rate not less than 1,400 shots per minute (spm), and shall function in the M61 gun at a cyclic rate not less than 5500 spm.

3.16 Function and casualty.-The cartridge, at ambient temperature, shall function without casualty.

3.17 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications, and the following:

3.17.1 Process defects.-The cartridge shall be free of defects such as folds, wrinkles, buckles, deep draw scratches, cuts, dents, perforations, and other forms of substandard workmanship.

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3.17.2 Cleanliness.-The cartridge and its components shall be free of all foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance terms and definitions.-Reference shall be made to MIL-STD-109 for definition of quality assurance terms.

4.2 Manufacturing process.-Submission of the proposed process and subsequent changes for approval, shall be as directed by the contracting officer.

4.3 First article sample.

4.3.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of:

a. 300 cartridges for physical and dimensional inspection and non-firing tests.

b. 585 cartridges for firing tests.

c. 32 projectiles (charged) for drop sensitivity tests.

The sample shall be manufactured using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

4.3.1.1 Examination and test.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a government laboratory or other such facility specified in the contract. (See 6.2)

4.3.1.2 Initial production sample failure.-Failure of the sample to comply with requirements of the drawings and specifications shall result in sample disapproval.

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4.4 Inspection provisions.

4.4.1 Submission of product.-The product shall be submitted in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.4.1.1 Lot.-A lot shall consist of:

a. Cartridges loaded by one manufacturer, in one unchanged process, in accordance with the same drawings and drawing revisions, and same specification and specification revision.

b. Like parts and assemblies (cases; primers; projectiles) having one manufacturer's symbol and one interfix number.

c. Propellant from one lot.

4.4.1.2 Lot identification.-Each cartridge and each packed ammunition lot shall be identified in accordance with applicable drawings and MIL-STD-1168, supplemented as directed by the procuring agency. Each packed lot shall be further identified by a federal stock number assigned by the procuring agency.

4.4.2 Examination.-One hundred percent examination shall be performed by the contractor for all critical defects. Examination for major and minor defects shall be performed as specified herein. Any cartridge, or any linked belt (when applicable), found to be defective, shall be rejected.

4.4.2.1 Primed cartridge cases.-Each primed cartridge case shall be tested by the contractor for electrical resistance subsequent to primer crimping and waterproofing, using the method of 4.5.2.1. Primed cases which fail to meet the requirements of 3.5 shall be removed from the lot.

4.4.2.2 Sampling and acceptance criteria.-Sampling plans and procedures shall be in accordance with MIL-STD-105 or MIL-STD-1235, except as otherwise indicated. Defect characteristics may be grouped into a class and considered collectively, or defect characteristics may be considered individually to determine acceptability. When acceptability is determined on a class basis, in addition to the AQL assigned to the major class, an AQL shall be assigned to individual characteristics within the major class, limiting the number of defectives containing a particular major defect to less than the number of defectives permitted for the class. However, when three or less defect characteristics are listed in a classification or when sampling procedures of MIL-STD-1235 are used, acceptability shall not be determined on a class basis for major defects. (See 6.4)

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4.4.2.3 AQL's.-The following AQL's are assigned.

a. Individual defect characteristic basis.

Major defects ----------0.25%
Minor defects ----------0.40%

b. Defect characteristic, class basis.

Major defects ----------1.50%
Minor defects ----------2.50%

c. Individual defect characteristic associated with the class basis.

Major defects ----------0.40%

4.4.3 Classification of defects 1/.-The classification of defects shall be as follows (see 6.5):

4.4.3.1 Nose, charged assembly (see Dwg. C7258806, a detail of Dwg. D7259219).

Category and defects	Method of inspection
Critical: None defined.	
Major:	
101. Length, shoulder to top of charge, min.	Gage
Minor: None defined.	

4.4.3.2 Projectile, 20MM, API-T, M52E1 (see Dwg. D7259219, a detail of Dwg. D7259218).

Category and defects	Method of inspection
Critical: None defined.	
Major:	
101. Depth to metering disc	Gage
102. Metering disc missing	Visual
Minor:	
201. Seal missing or perforated	Visual
202. Gap between nose and adapter	Gage
203. Concentricity, nose with bourrelet	Gage

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4.4.3. Cartridge assembly (see Dwg. D7259218).

Category and defects	Method of inspection
Critical:	
1. Crack or split in projectile assembly	Visual
2. Split or perforated case	Visual
Major:	
101. Loose primer	Manual
102. Depth of primer seating	Gage
103. Over-all length, max	Gage
104. Improper or missing crimp (case-projectile)	Visual
105. Profile and alignment, max	Gage
106. Corrosion	Visual
107. Head configuration	Visual
108. Metal sliver(s) across primer insulator	Visual
109. Length, case head to centerline of crimp	Gage
Minor:	
201. Marking incorrect, illegible or missing	Visual
202. Protective coating damaged or incomplete	Visual
203. Waterproofing missing around primer	Visual
204. Scratch, dent, bulge, wrinkle or fold in case	Visual
205. Rotating band damaged	Visual
206. Lacquer on primer button	Visual
207. Primed crimp missing	Visual
208. Incorrect type cartridge	Visual
209. Foreign matter except corrosion	Visual (see 6.6

1/ MIL-STD-651 shall apply in defining and evaluating cartridge visual defects.

4.4.4 Tests.-The tests listed in Table I shall be performed on each lot in accordance with the test methods prescribed in 4.5. For acceptance, each test shall comply with the applicable requirement(s).

4.4.4.1 Test samples.-The quantities for the various tests shall be as specified in Table I. Except as otherwise specified, tests shall be conducted with samples at ambient temperature.

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Table I

Test		Sample Size Production Lot		Rqmt Para
Projectile extraction	1/	20		3.4
Electrical resistance verification	2/	20	16/	3.5
Propellant contamination	3/	20	16/	3.17.2
Projectile torque	4/	20		3.6
Velocity	5/	20		3.7
Pressue	6/	20	16/	3.8
Waterproofness	7/	20		3.9
Accuracy	8/	50		3.10
Action time	9/			3.11
Ambient temperature		25	16/	
-65°F. temperature		25		
Incendiary performance	10/	20		3.12
Drop sensitivity	11/	32		3.13
Projectile trace	12/	50		3.14
Cyclic rate				3.15
M39 gun	13/	200	16/	
M61 gun	14/	200	16/	
Function and casualty	15/			3.16
M39 gun		200		
M61 gun		200		

1/ Projectile extraction.-Failure of two or more sample cartridges to comply with the minimum requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply with the minimum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if 2 or more cartridges of the combined sample fails to comply with the minimum requirement. Failure of 4 or more cartridges to comply with the maximum requirement shall cause rejection of the lot. If more than 1 but less than 4 cartridges fail to comply with the maximum requirement, a second sample of 20 cartridges shall be tested. The lot shall be rejected if 5 or more cartridges of the combined sample fail to comply with the maximum requirement.

2/ Electrical resistance verification.-The lot will be rejected if any primer of the sample cartridges fails to comply with either or both of the primer resistance requirements.

3/ Propellant contamination.-The presence of foreign matter in the propellant of any of the test cartridges will cause rejection of the lot (see 3.17.2 and 6.6).

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4/ Projectile torque.-Failure of 2 or more of the sample cartridges to comply with the specified torque requirement shall cause rejection of the lot. If one cartridge of the sample fails to comply, a second sample of 20 cartridges shall be tested. The lot shall be rejected if 2 or more cartridges of the combined sample fails to comply with the applicable requirement.

5/ Velocity.-If the sample fails to comply with either or both requirements, a second sample of 40 cartridges shall be tested for the failing parameter(s) (average, sample standard deviation). The lot shall be rejected if the second sample fails to comply with the applicable requirements (see 6.7.1).

6/ Pressure.-If the average pressure ($\bar{X}$) plus 0.84 times the average range ($\bar{R}$) of the sample cartridges exceeds 60,500 psi, a second sample of 40 cartridges shall be tested. The lot shall be rejected if the average pressure ($\bar{X}$) plus 0.91 times the average range ($\bar{R}$) of the second sample fails to comply with the applicable requirement (see 6.7.2).

7/ Waterproofness.-If the average velocity (wet) of the sample fails to comply with the applicable requirement, a second sample of 40 cartridges shall be tested. The lot shall be rejected if the average velocity (wet) differs by more than 100 fps from the average velocity (dry) of 40 additional sample cartridges fired to establish a new base for comparison. The lot shall not be penalized for failure of the last named sample cartridges to comply with the requirement of 3.8.

8/ Accuracy.-If the average mean radius of the sample cartridges exceeds the applicable requirement, a second sample of 100 cartridges shall be tested. The lot shall be rejected if the average mean radius of ten accuracy targets exceeds 15 inches at the same range.

9/ Action time.-For acceptance, each cartridge of the test sample shall exhibit an action time not exceeding 4 milliseconds (0.004 second) and the sample average action time ($\bar{X}$) plus 4 sample standard deviations (S) shall not exceed 4 milliseconds. These criteria apply separately to both the ambient and cold phase tests.

10/ Incendiary performance.-Failure of three or more sample cartridges to comply with the requirement shall cause rejection of the lot. If one or two cartridges of the sample fail to comply with the requirement, a second sample of 20 cartridges shall be tested. If in the combined first and second samples four or more cartridges fail to comply with the requirements, the lot shall be rejected.

11/ Drop sensitivity.-The functioning of a projectile of the sample upon impact against the plate shall cause rejection of the lot.

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12/ Projectile trace.-Failure of nine or more sample cartridges to comply with the requirement shall cause rejection of the lot. If more than five and less nine of the sample cartridges fail to comply with the requirement, a second sample of fifty cartridges shall be tested. If in the combined first and second samples thirteen or more sample cartridges fail to comply with the requirement, the lot shall be rejected.

13/ Cyclic rate, M39 gun.-A test weapon which fails to operate at the specified cyclic rate with any burst of test cartridges shall be tested with a burst of reference cartridges. If the reference cartridges operate the test weapon at the specified rate, indicative of test cartridge failure, the lot shall be rejected. If the burst of reference cartridges fails to operate the test weapon at the specified rate, indicative of weapon failure, the procedure of 4.5.13 shall apply.

14/ Cyclic rate, M61 gun.-If the cyclic rate of the M61 gun falls below 5500 spm, the test shall be suspended and all test results, except for the occurrence of other firing defects correctly chargeable against the ammunition, shall be nullified. The test shall be continued after correction or replacement of the failing gun.

15/ Function and casualty.-The lot shall be rejected if any malfunction or firing casualty of Table II occurs in number(s) equal to or greater than the applicable "Rej" number. Except as otherwise provided, if malfunctions or casualties occur in excess of the applicable "Acc" number, but less than the applicable "Rej" number, a second sample of 800 cartridges shall be divided and fired in equal numbers in the M39 and M61 guns. The lot shall be rejected if in the combined samples, malfunctions or casualties occur in numbers equal to or greater than the applicable "Rej" number.

16/ Additional cartridges are not required for these tests as cartridges from other test samples may be used as follows:

a. Primed cases and propellant from the projectile extraction test are used as samples for the electrical resistance verification and propellant contamination tests.

b. Velocity and pressure tests are fired concurrently.

c. Action time, ambient temperature phase, test is conducted concurrently with the accuracy test.

d. Cyclic rate is determined in the function and casualty test.

4.4.4.2 Firing defects.-Firing defects and acceptance criteria shall be as specified in Table II. For defect definitions, see AMCR 715-505, Volume 8.

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Table II

Firing Casualties	Acc	Rej
Cartridge:		
Fire "out of battery," M61 gun 1/	0	1
Misfire 2/	-	-
Failure to chamber	0	1
Failure to extract	0	1
Projectile remaining in bore	0	1
Primer:		
Missing primer, or button missing from primer	0	2
Primer leak	15	40
Case:		
Longitudinal split 3/		
I or S	10	31
J	4	11
K	0	2
L or M	0	1
Circumferential rupture, partial 3/		
S, J or K	4	11
L	0	1
Circumferential rupture, complete 3/	0	1
Detached metal 4/	0	1
Projectile:		
Rotating band separation 5/		
Complete or partial separation having band seat serration imprint	0	3
Separation having length greater than 0.5 inch, and without band seat serration imprint	1	6
Metal parts separation, except rotating band separation 6/	0	1
Premature functioning	0	1

1/ A cartridge shall be classed defective if it fires, due to delayed ignition, after the bolt has been unlocked and extraction has been initiated or completed.

2/ Misfiring test cartridges, excepting those from the waterproofness test (Table I), shall be retested in accordance with the method of 4.4.10. If a cartridge fails to function in that test, examinations shall be conducted for the following conditions:

a. Excessive primer seating depth.
b. Metal slivers across primer insulator.

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c. Electrical continuity.
d. Missing or obstructed case vent.
e. Missing or insufficient primer composition or propellant.
f. Partially burned or contaminated primer composition or propellant.

The lot shall not be penalized if condition a. is observed. However, this condition shall be reported for information. The "Acc" number shall be zero (0) and the "Rej" number 3, for misfires associated with conditions b., c., or d. The second sample for evaluation of misfires associated with conditions b., c., or d. shall consist of the function and casualty second sample (see 4.3.3.12). The lot shall be rejected if either condition e. or f. is observed. Misfire(s) occurring in the waterproofness test shall be handled in accordance with AMCR 715-505, Volume 8.

3/ See Dwg. C7258801 for classifying splits and ruptures in fired cartridge cases. If a longitudinal split or circumferential rupture (partial) extends into two or more defined areas, only the most severe defect criterion of Table II for the areas involved, shall apply. If a rupture results in separation of the cartridge case into two or more portions, the defect shall be classed as a complete circumferential rupture.

4/ Metal sheared or missing from the fired cartridge case exterior, such as rim or neck shears, shall be classed as a defect. The lot shall not be penalized for shavings of metal from the interior wall of the case neck, in the crimped area.

5/ For classification as a defect, there must be evidence either by recovery of the band, or portion thereof, or by hole(s) in the fragmentation screen(s). The lot shall not be penalized for normal band fringing, or the separation of band fragments having no evidence of band seat serration imprint and having length of 0.5 inch or less.

6/ Separation or breakup of projectile part(s), except rotating band separation, as evidenced by recovery of the part(s) or fragment(s), or hole(s) in the fragmentation screen(s), shall be classed as a defect.

4.4.4.3 Unlisted firing defects.-The lot shall be suspended and referred to the contracting officer for disposition, if a malfunction or casualty not covered by this specification occurring in any firing test, indicates that the product is unsuited for the purpose intended.

4.4.5 Packaging, packing and marking inspection.-Inspection for packaging, packing and marking shall be in accordance with MIL-STD-644, as applicable to the drawing.

4.4.6 Inspection equipment.-Inspection equipment conforming to the designs on applicable drawings listed by IEL 7259218 shall be used to perform the specified examinations and tests.

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4.5 Test methods and procedures.

4.5.1 Projectile extraction.-The method of test shall be as prescribed in AMCR 715-505, Volume 8.

4.5.2 Electrical resistance.

4.5.2.1 Primed cartridge cases.-The equipment and circuitry specified in IEL 7259218 shall be used to measure the resistance of the primer in each primed cartridge case.

4.5.2.2 Verification.-A Simpson Volt-Ohm-Ammeter, Model 260, or approved equal, will be used to measure the resistance of the primer in each primed cartridge case from the projectile extraction test sample. Only the RX 100 and RX 10,000 scales of the meter will be used in order to limit the applied voltage and current, respectively, to 7.5 volts and 1.4 milliamperes, maximum. This test will be conducted by Government personnel.

4.5.3 Propellant contamination.-The propellant from each of the cartridges tested for projectile extraction will be examined visually by Government personnel for contamination, as prescribed in AMCR 715-505, Volume 8.

4.5.4 Projectile torque.-Equipment shall be as specified in IEL 7259218. Each cartridge of the test sample shall be marked with a light scratch extending axially across the rotating band onto the case neck. Torque to 10 inch pounds shall be applied slowly to the projectile. Movement of the projectile with respect to the cartridge case, detectable by misalignment in the scratch mark, shall be recorded.

4.5.5 Velocity, pressure and waterproofness.-The methods of test shall be as prescribed in AMCR 715-505, Volume 8. Statistics necessary for testing of results against acceptance criteria shall be computed (see 6.7).

4.5.6 Accuracy.-The method of test and measurement of targets shall be as prescribed in AMCR 715-505, Volume 8. The specified range may be shortened in decrements of 100 yards, to a minimum range of 200 yards. For ranges less than 600 yards the average mean radius shall be reduced in direct proportion to the reduction in range.

4.5.7 Action time.-The method of test shall be as prescribed in AMCR 715-505, Volume 8. Statistics necessary for testing of results against the acceptance criterion shall be computed (see 6.7).

4.5.8 Incendiary performance.-The method of test and grading of incendiary performance shall be as prescribed in AMCR 715-505, Volume 8. The target shall be located 200 ± 25 yards from the gun muzzle, normal to the line of fire.

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4.5.9 Drop sensitivity.-The method of test shall be as prescribed in AMCR 715-505, Volume 8. The nose of the projectile shall impact the steel plate, normal to projectile axis at a velocity of 52 ± 5 fps.

4.5.10 Projectile trace.-The method of test shall be as prescribed in AMCR 715-505, Volume 8. Visual observation(s) or photographic witness of trace characteristics shall be made from suitable position(s) approximately 90° from line of fire.

4.5.11 Function and casualty.-The method of test shall be as prescribed in AMCR 715-505, Volume 8. The test sample shall be fired in bursts of 50 cartridges. There shall be observation for a distance of 600 feet from the gun muzzle for premature bursts. The gun barrel(s) shall be at ambient temperature for the firing of each burst.

4.5.12 Misfires.-Misfiring cartridges shall be retested in rifle as prescribed in AMCR 715-505, Volume 8, for the accuracy test. Results of examinations required by Table II shall be recorded.

4.5.13 Defective weapon.-If any test cartridge firing defect is found to have been caused by a defective test weapon, the firing defect shall not be valid for lot penalty. In such case, the defective test weapon shall be corrected or replaced and the test repeated in whole or in part, as indicated. If such firing defect is not found to have been caused by the defective weapon, it shall be valid for lot penalty.

4.5.14 Defect penalty.-In any ballistic test, except function and casualty, in which the occurrence of a firing defect listed in Table II prevents the obtaining of a reliable result for the characteristic being tested, except as otherwise specified, an additional shot shall be fired. That particular test shall not be penalized, but the total ballistic sample shall be penalized for such defects in accordance with Table II.

5. PREPARATION FOR DELIVERY

5.1 Packing, Level A.-The type of ammunition pack shall be in accordance with instructions from the procuring agency. Linked ammunition shall be packed in accordance with Dwg. F7258968, F7258969, or F7258970. Bulk ammunition shall be packed in accordance with Dwg. F7258941 or F7258942.

5.2 Marking.-Marking on the packed container shall be in accordance with Dwg. D7258928 or D7259237, as applicable.

5.3 Data cards.-Ammunition data cards shall be furnished. The requirements of MIL-STD-1167 shall apply.

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6. NOTES

6.1 Ordering data.-Invitation for bids and contracts or orders will specify the following:

6.1.1 Title, number, and date of this specification.

6.1.2 Place of inspection, if not at place of manufacture.

6.1.3 First article sample requirements (see 3.2).

6.1.4 Detailed packing instructions (see 5.1).

6.1.5 Coded markings required by Dwg. D7258928 (see 5.2).

6.1.6 Provision for the supply, maintenance and disposition of mandatory ballistic test equipment for acceptance inspection purposes.

6.2 First article sample.-The procurement agency may waive the requirement for a first article sample if the contractor has recently demonstrated his ability to produce this item. The contracting officer will notify the testing facility when to proceed with the firing tests.

6.3 Process deviation.-A process deviation is defined as a change in the approved basic method of manufacture, or an operational change which may alter the metallurgical or physical properties of the item.

6.4 AQL's.-The optional use of AQL values for either individual defects or classes of defects, with individual major defect limitation, is intended to minimize inspection agency administrative burden which might result from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the manufacturing process.

6.5 Intermediate point inspection.-The classification of defects identifies the defect characteristics for acceptance inspection. It may be necessary to modify the sequence of inspection stations to best suit the manufacturing process. Inspection for defect characteristics which will be hidden or altered by subsequent processing operations (including unrelated operations), should be scheduled to prevent premature acceptance which could be detrimental to the attainment of optimum product quality of the item.

6.6 Contamination of explosive components.-Particular care should be exercised to avoid contamination of primers and propellant by oil, grease, or other foreign matter since such contamination can cause misfire(s) or increase hangfire time to make the ammunition unsuitable for aircraft use.

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6.7 Computations.

6.7.1 Standard deviation.-Where computation of a sample standard deviation is specified for determination of lot acceptance, any standard method of computation may be used.

6.7.2 Range plan.-Where a range plan is specified, the individual values will be recorded in subgroups of 5 in the order in which they are observed. The ranges for the subgroups will be added and the sum of the ranges will be divided by the number of subgroups to determine the average range ($\bar{R}$).

Custodian:

Army - MU

Preparing Activity:

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